DIMENSION OF A PRODUCT OF LOCALLY ALGEBRAIC SCHEMES IS THE SUM OF THE DIMENSIONS

THE STACKS PROJECT AUTHORS

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Editorial note. The label lemma-dimension-product-locally-algebraic is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Flatness is preserved by base change. Morphisms, Lemma morphisms-lemma-base-change-flat, invoked at proof lines 4022-4024. Supplies flatness of the projection to Y.
- Dimension of a fibre after base change. Morphisms, Lemma morphisms-lemma-dimension-fibre-after-base-change, invoked at proof lines 4025-4026. Supplies the identification of the fibre dimension at the point with the dimension of X there.
- Additivity of fibre dimension along a flat morphism of locally algebraic schemes. Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point-additive, invoked at proof lines 4027-4028. Supplies the sum formula at the point.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02JS: sources/stacks-chow/tags.txt, source lines 3347–3347. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *The dimension of the product is the sum of the dimensions. Let k be a field. Let X, Y be locally algebraic k-schemes.*

(1) *For $z \in X \times Y$ lying over (x, y) we have $\dim_z(X \times Y) = \dim_x(X) + \dim_y(Y)$.*
(2) *We have $\dim(X \times Y) = \dim(X) + \dim(Y)$.*

Proof. Proof of (1). Consider the factorization

$$X \times Y \longrightarrow Y \longrightarrow \mathrm{Spec}(k)$$

of the structure morphism. The first morphism $p : X \times Y \to Y$ is flat as a base change of the flat morphism $X \to \mathrm{Spec}(k)$ by Morphisms, Lemma 01U9. Moreover, we have $\dim_z(p^{-1}(y)) = \dim_x(X)$ by Morphisms, Lemma 02FY. Hence $\dim_z(X \times Y) = \dim_x(X) + \dim_y(Y)$ by Morphisms, Lemma 02JS. Part (2) is a direct consequence of (1). □